EXAM SHIELD

Question Paper

Computing question bank: 11 subjects for proctored exams

110 questions: 88 multiple choice and 22 written. Total 174 marks.
Edition: question paper (no answers shown).

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How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Golang

Concurrency, error handling, types and tooling in Go.

1. Which keyword starts a new goroutine?

A. go
B. spawn
C. async
D. thread

2. What is the value of a map declared with var m map[string]int before it is initialised?

A. A map with one zero entry
B. An empty map ready for writes
C. It does not compile
D. nil

3. What happens when a program writes to a nil map in Go?

A. The map is created automatically
B. The write is silently ignored
C. A compile-time error
D. A runtime panic

4. Which statement schedules a function call to run when the surrounding function returns?

A. later
B. defer
C. await
D. finally

5. How does idiomatic Go report a recoverable failure from a function?

A. By throwing an exception
B. By returning an error value as the last result
C. By setting a global errno variable
D. By calling panic in every case

6. Which built-in function creates a channel of integers?

A. channel(int)
B. new(chan int)
C. make(chan int)
D. chan.New(int)

7. What does a select statement do in Go?

A. Waits on several channel operations and runs the one that is ready
B. Chooses which goroutine the scheduler stops
C. Selects rows from a database table
D. Sorts the elements of a slice

8. Which command formats Go source code to the standard style?

A. go style
B. gopretty
C. gofmt
D. golint

9. Explain the difference between an array and a slice in Go, including how each is passed to a function. [4 marks]

10. What is a data race? Describe how the Go race detector helps you find one. [4 marks]

Section 2: Python

Core syntax, data types and idioms of Python 3.

11. Which of these Python types is immutable?

A. dict
B. tuple
C. set
D. list

12. Which keyword defines a function in Python?

A. def
B. func
C. fn
D. function

13. What does len("hello") return?

A. 6
B. 4
C. An error
D. 5

14. What is the result of the expression 7 // 2 in Python 3?

A. 1
B. 3.5
C. 3
D. 4

15. Which line is a valid dictionary comprehension?

A. [k: v for k, v in pairs]
B. (k: v for k, v in pairs)
C. {k: v for k, v in pairs}
D. dict[k for k in pairs]

16. What does a with open(path) as fh: block guarantee?

A. The file is closed when the block exits
B. The file is locked for other processes
C. The file is deleted afterwards
D. The file is opened read-only

17. Which tool installs third-party Python packages?

A. npm
B. gem
C. composer
D. pip

18. What does print(type([])) display?

A. <class 'list'>
B. <class 'dict'>
C. <class 'array'>
D. <class 'tuple'>

19. Explain the Global Interpreter Lock (GIL) and its effect on multi-threaded, CPU-bound Python code. [4 marks]

20. Compare a list with a generator. When is a generator the better choice? [4 marks]

Section 3: JavaScript

Language fundamentals, types and asynchronous programming.

21. Which keyword declares a block-scoped variable that cannot be reassigned?

A. var
B. let
C. const
D. static

22. What does typeof null evaluate to?

A. "null"
B. "object"
C. "number"
D. "undefined"

23. What is the result of "5" + 3 in JavaScript?

A. "53"
B. A TypeError
C. 8
D. NaN

24. Which operator compares both value and type without coercion?

A. =>
B. ==
C. =
D. ===

25. Which method turns a JSON string into a JavaScript object?

A. JSON.toObject
B. Object.fromJSON
C. JSON.parse
D. JSON.stringify

26. Which syntax, standardised in ES2017, lets you write promise-based code that reads synchronously?

A. async / await
B. XMLHttpRequest
C. Generators with yield
D. Callbacks

27. Which array method returns a new array containing each element transformed by a function?

A. push
B. splice
C. map
D. forEach

28. When does a setTimeout callback with a delay of 0 ms run?

A. After the current call stack and pending microtasks finish

B. Immediately, before the next line of code

C. Never; a delay of 0 is ignored

D. Before any pending promise callbacks

29. Explain what a closure is and give one practical use case. [4 marks]

30. Describe the differences between var, let and const. [3 marks]

Section 4: PHP

Server-side PHP: syntax, forms, security and tooling.

31. Which character begins every variable name in PHP?

A. @
B. $
C. #
D. &

32. Which superglobal holds data submitted by an HTML form using the POST method?

A. $_POST
B. $_FORM
C. $_SERVER
D. $_GET

33. Which operator joins two strings in PHP?

A. . (dot)
B. ++ (increment)
C. + (plus)
D. & (ampersand)

34. Which function checks that a variable exists and is not null?

A. isset()
B. is_null()
C. empty()
D. defined()

35. What is the correct way to prevent SQL injection when using PDO?

A. Calling addslashes() on input
B. Running strip_tags() on input
C. Hashing the input with md5()
D. Prepared statements with bound parameters

36. Which tool manages PHP project dependencies?

A. npm
B. pip
C. Maven
D. Composer

37. Which function should be used to hash user passwords?

A. password_hash()
B. md5()
C. base64_encode()
D. sha1()

38. What does echo 10 <=> 5; print?

A. 1

B. 0

C. -1

D. true

39. Explain the difference between include and require in PHP. [3 marks]

40. What are PHP sessions? List two measures that keep session data secure. [4 marks]

Section 5: Laravel

The Laravel framework: routing, Eloquent, Blade and Artisan.

41. What is the name of Laravel's command-line tool?

A. Tinker
B. Artisan
C. Forge
D. Blade

42. Which templating engine ships with Laravel?

A. Twig
B. Smarty
C. Blade
D. Mustache

43. What is Laravel's built-in ORM called?

A. Sequelize
B. Hibernate
C. Eloquent
D. Doctrine

44. Which command creates a new migration file?

A. php artisan make:migration create_posts_table
B. php artisan db:create posts
C. php artisan migrate:new posts
D. composer make:migration posts

45. In which file are browser (web) routes normally defined?

A. app/routes.php
B. routes/web.php
C. config/routes.php
D. public/routes.php

46. What is the main purpose of middleware in Laravel?

A. Defining the database schema
B. Filtering HTTP requests entering the application
C. Compiling front-end assets
D. Rendering Blade views

47. Which Blade directive adds a CSRF token field to a form?

A. @protect
B. @secure
C. @csrf
D. @token

48. Which command applies all pending database migrations?

A. php artisan db:run
B. composer migrate
C. php artisan migrate
D. php artisan schema:apply

49. Explain mass assignment and how the $fillable property protects against it. [4 marks]

50. Describe Laravel's service container and how it supports dependency injection. [4 marks]

Section 6: Computer Science

Algorithms, data structures, number systems and systems.

51. What is the time complexity of binary search on a sorted array?

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)

52. Which data structure follows Last-In, First-Out (LIFO) order?

A. Queue
B. Linked list
C. Binary search tree
D. Stack

53. What is the average-case time complexity of quicksort?

A. O(n log n)
B. O(n^2)
C. O(n)
D. O(log n)

54. What is the binary representation of decimal 10?

A. 1100
B. 0110
C. 1001
D. 1010

55. Which algorithm finds shortest paths in a graph with non-negative edge weights?

A. Depth-first search
B. Dijkstra's algorithm
C. Kruskal's algorithm
D. Prim's algorithm

56. What does a compiler do?

A. Translates source code into machine or target code
B. Executes source code one line at a time
C. Compresses files for storage
D. Translates machine code back into source code

57. What is the average time to look up a key in a hash table?

A. O(1)
B. O(n^2)
C. O(n)
D. O(log n)

58. Which of these is NOT a core operating system function?

A. Compiling user programs

B. Process scheduling

C. Memory management

D. File management

59. Explain recursion and the role of the base case. Give a simple example. [3 marks]

60. Compare a stack and a queue, giving one real-world use for each. [4 marks]

Section 7: Information Technology

Networking, infrastructure, cloud services and IT operations.

61. Which protocol automatically assigns IP addresses to devices on a network?

A. FTP
B. DNS
C. SMTP
D. DHCP

62. What is the main job of DNS?

A. Assigning MAC addresses
B. Translating domain names into IP addresses
C. Routing email between servers
D. Encrypting web traffic

63. What is the default port for HTTPS?

A. 80
B. 443
C. 21
D. 8080

64. Which device forwards traffic between different networks?

A. Repeater
B. Router
C. Hub
D. Switch

65. What does RAID 1 provide?

A. Distributed parity across three disks
B. Striping without redundancy
C. Mirroring of data across disks
D. Data compression

66. Which of these is a cloud service model?

A. LAN (Local Area Network)
B. VPN (Virtual Private Network)
C. SaaS (Software as a Service)
D. DHCP (Dynamic Host Configuration Protocol)

67. How many bits are in an IPv4 address?

A. 32
B. 64
C. 128
D. 16

68. What does the 3-2-1 backup rule recommend?

A. 3 servers, 2 administrators and 1 user

B. Daily, weekly and monthly backups only

C. 3 copies of data, on 2 types of media, with 1 copy offsite

D. 3 disks in a RAID 2 array plus 1 spare

69. Explain the difference between a LAN and a WAN, with one example of each. [3 marks]

70. Describe the role of an IT help desk and how a ticketing system supports it. [4 marks]

Section 8: Information Systems

How organisations use data, systems and processes.

71. What does TPS stand for in information systems?

A. Team Performance System
B. Total Planning Service
C. Text Processing Software
D. Transaction Processing System

72. What does an ERP system do?

A. Handles only accounting
B. Hosts the company website
C. Manages only email
D. Integrates core business processes across departments

73. How does information differ from data?

A. There is no difference
B. Information is raw, unprocessed facts
C. Information is data processed into a meaningful form
D. Information refers to computer hardware

74. What is a Decision Support System (DSS) mainly used for?

A. Printing documents
B. Running payroll only
C. Helping managers analyse options and make decisions
D. Routing network traffic

75. What is the main goal of database normalisation?

A. Increasing hardware speed
B. Adding more users
C. Encrypting the database
D. Reducing data redundancy

76. Which SDLC phase identifies what users need from a new system?

A. Deployment
B. Maintenance
C. Testing
D. Requirements analysis

77. What does a CRM system focus on?

A. Managing CPU scheduling
B. Managing legal compliance
C. Managing source code
D. Managing relationships with customers

78. What does a primary key do in a relational table?

A. Encrypts the table

B. Sorts rows alphabetically

C. Uniquely identifies each row

D. Links to a table in another database only

79. Explain how a Management Information System (MIS) supports middle management. [4 marks]

80. Discuss two ethical issues that arise from the use of information systems. [4 marks]

Section 9: Cyber Security

Security principles, threats, cryptography and defence.

81. What do the letters of the CIA triad stand for?

A. Cipher, Identity, Access
B. Control, Inspection, Audit
C. Compliance, Integrity, Authentication
D. Confidentiality, Integrity, Availability

82. What is phishing?

A. Guessing passwords by trying every combination
B. Tricking people into revealing information through fake messages
C. Capturing packets on a network
D. Flooding a server with requests

83. Which of these is an asymmetric (public-key) algorithm?

A. AES
B. DES
C. 3DES
D. RSA

84. What is the most effective defence against SQL injection?

A. Longer passwords
B. Parameterised queries
C. Using HTTPS
D. Installing a firewall

85. What does multi-factor authentication require?

A. Two or more independent proofs of identity
B. Two or more administrator accounts
C. Two or more firewalls
D. Two or more backups

86. Which property describes a cryptographic hash function?

A. It is one-way and cannot practically be reversed
B. It compresses files losslessly
C. It can be decrypted with the right key
D. It is the same as Base64 encoding

87. A DDoS attack mainly threatens which security goal?

A. Confidentiality
B. Integrity
C. Non-repudiation
D. Availability

88. What does the principle of least privilege mean?

A. Every user is an administrator

B. All users have read-only access to everything

C. Passwords are not required

D. Users get only the access they need to do their work

89. Explain the difference between symmetric and asymmetric encryption, with an example of each. [4 marks]

90. Describe the main phases of an incident response plan. [5 marks]

Section 10: Software Engineering

Processes, design principles, testing and collaboration.

91. Which agile framework organises work into fixed-length sprints?

A. Spiral
B. Scrum
C. Waterfall
D. V-model

92. What does unit testing focus on?

A. Only the user interface
B. Only the hardware
C. The smallest testable parts of a program
D. The whole system end to end

93. Which Git command records staged changes in the repository history?

A. git push
B. git commit
C. git clone
D. git status

94. In the SOLID principles, what does the S stand for?

A. Simple design
B. Static typing
C. Single responsibility
D. Secure coding

95. What is refactoring?

A. Rewriting the program in another language
B. Adding new features
C. Improving code structure without changing its behaviour
D. Deleting failing tests

96. What does CI stand for in modern development practice?

A. Code Inspection
B. Central Index
C. Compiled Interface
D. Continuous Integration

97. Which UML diagram shows how objects interact over time?

A. Sequence diagram
B. Class diagram
C. Entity-relationship diagram
D. Deployment diagram

98. What is technical debt?

A. Unpaid developer salaries

B. Loans used to buy hardware

C. Money owed for software licences

D. The future cost of shortcuts taken in code today

99. Compare the Waterfall and Agile approaches to software development. [5 marks]

100. Explain the purpose of code reviews and two benefits they bring to a team. [3 marks]

Section 11: DevOps Engineering

Containers, orchestration, automation and observability.

101. In Docker, what is a container?

A. A hypervisor for virtual machines
B. A registry that stores images
C. A running instance of an image
D. A script that builds an image

102. Which file describes how to build a Docker image?

A. Dockerfile
B. Makefile
C. Jenkinsfile
D. package.json

103. What is the smallest deployable unit in Kubernetes?

A. Cluster
B. Pod
C. Namespace
D. Node

104. Which tool is used for Infrastructure as Code?

A. Terraform
B. Wireshark
C. Postman
D. Photoshop

105. What is a blue-green deployment?

A. A debugging technique
B. Running two identical environments and switching traffic between them
C. A code formatting standard
D. A colour scheme for dashboards

106. Which tool collects and stores time-series metrics for monitoring?

A. Webpack
B. Babel
C. Prometheus
D. Figma

107. In a CI/CD pipeline, what normally happens in the test stage?

A. Servers are physically installed
B. Automated tests run against the new build
C. Old builds are archived
D. The release is announced to customers

108. Which command lists the running Docker containers?

A. docker show
B. docker ps
C. docker ls-run
D. docker images

109. Explain what a CI/CD pipeline is and describe three benefits it brings. [5 marks]

110. Compare containers with virtual machines in terms of isolation, size and start-up time. [4 marks]